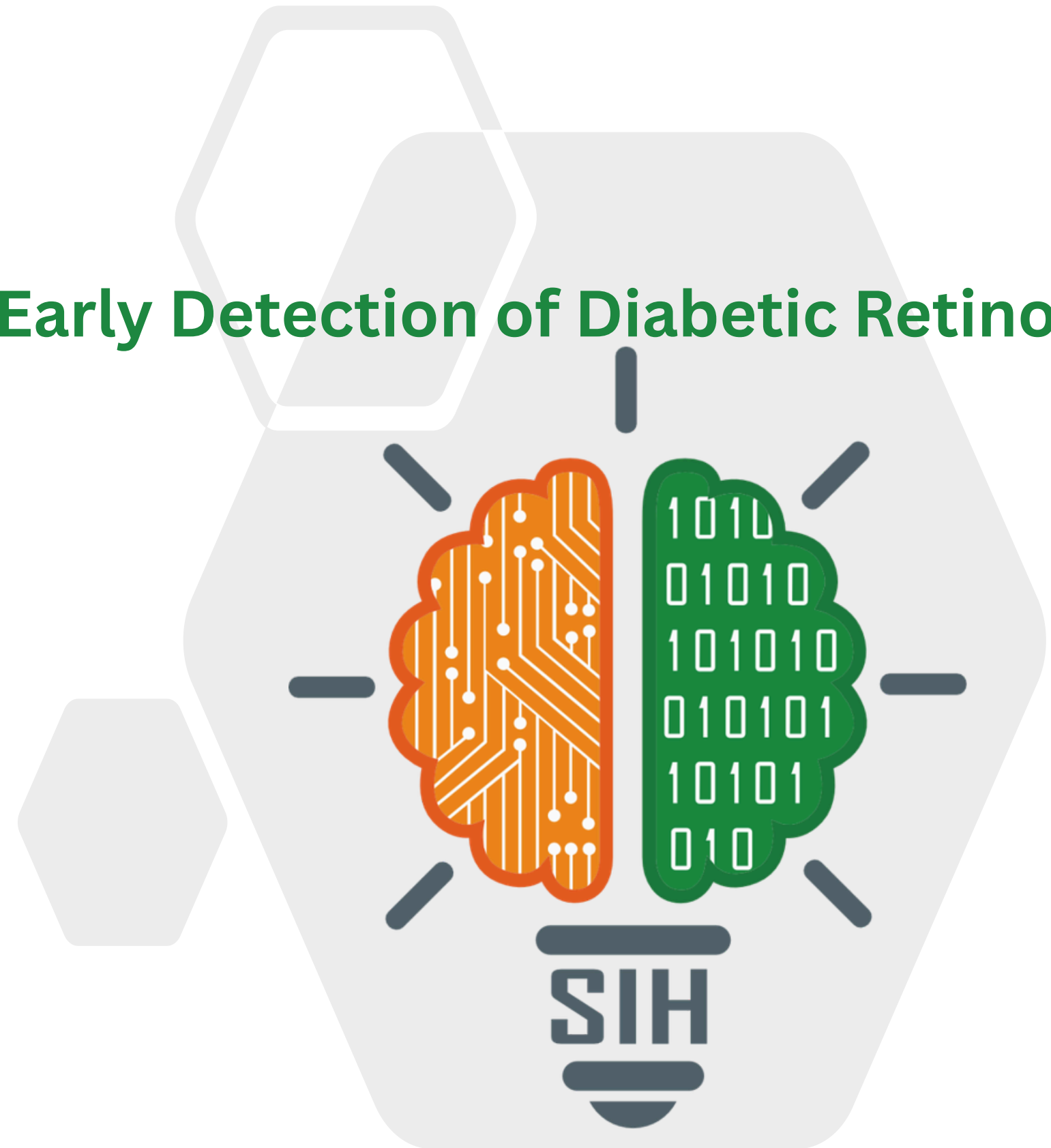


- Problem Statement ID - **SIH26038**
- Problem Statement Title- **AI-Powered Early Detection of Diabetic Retinopathy**
- Theme- **Miscellaneous**
- PS Category- **Software**
- Team ID-
- Team Name- **CodeStorm**



PROBLEM EXISTING

- [Poor-quality retinal images](#) : blur, low light, uneven illumination can cause wrong predictions.
- [Lack of ophthalmologists in rural areas](#) : manual screening is difficult to scale.
- [Image conditions](#) : AI trained on one dataset may perform poorly on images from different devices .

1 Poor-quality retinal images
Blur, low light, uneven illumination can cause wrong predictions.

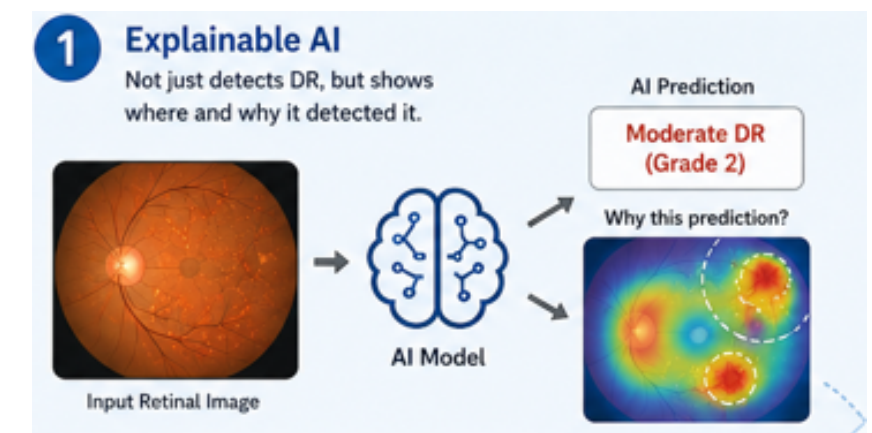


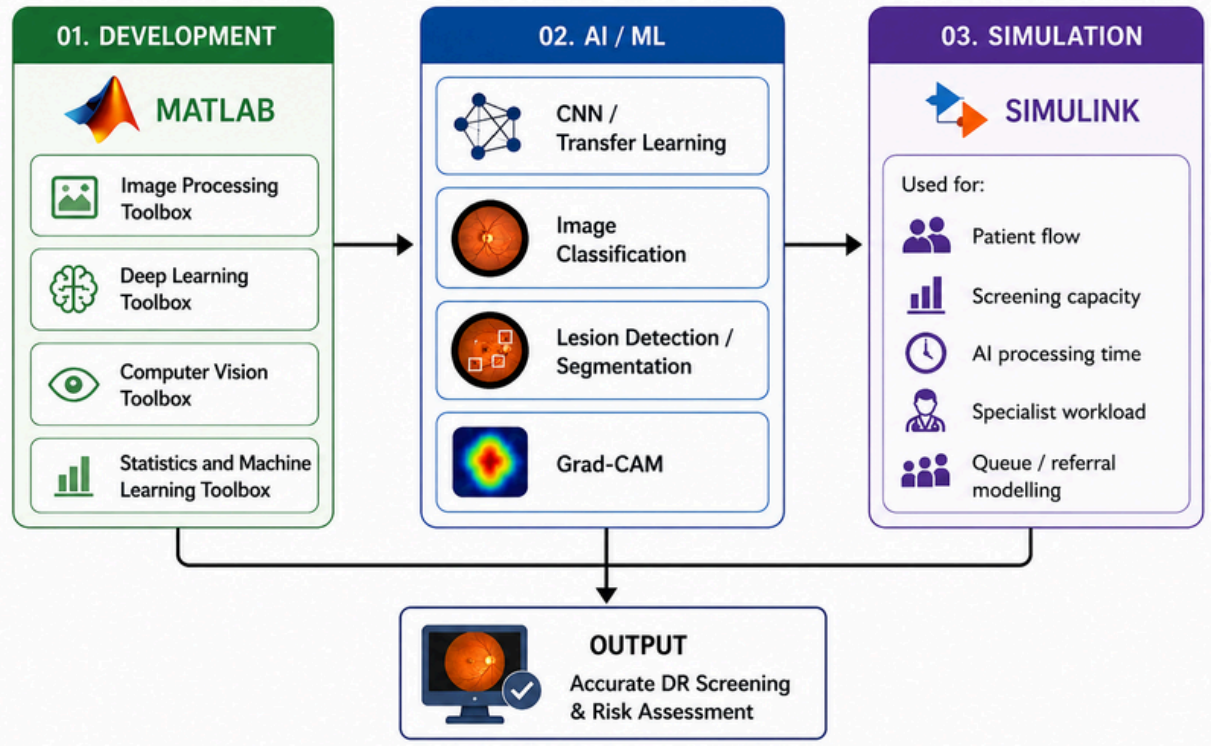
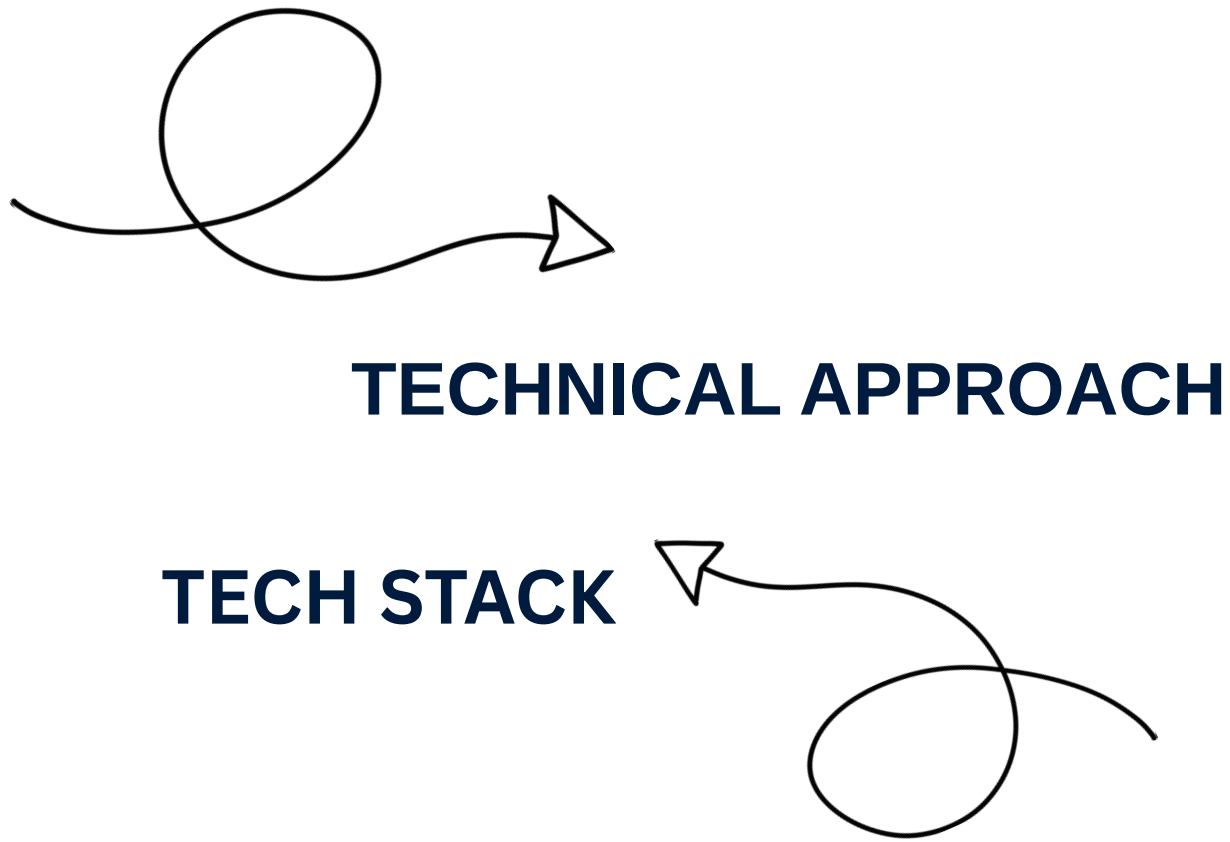
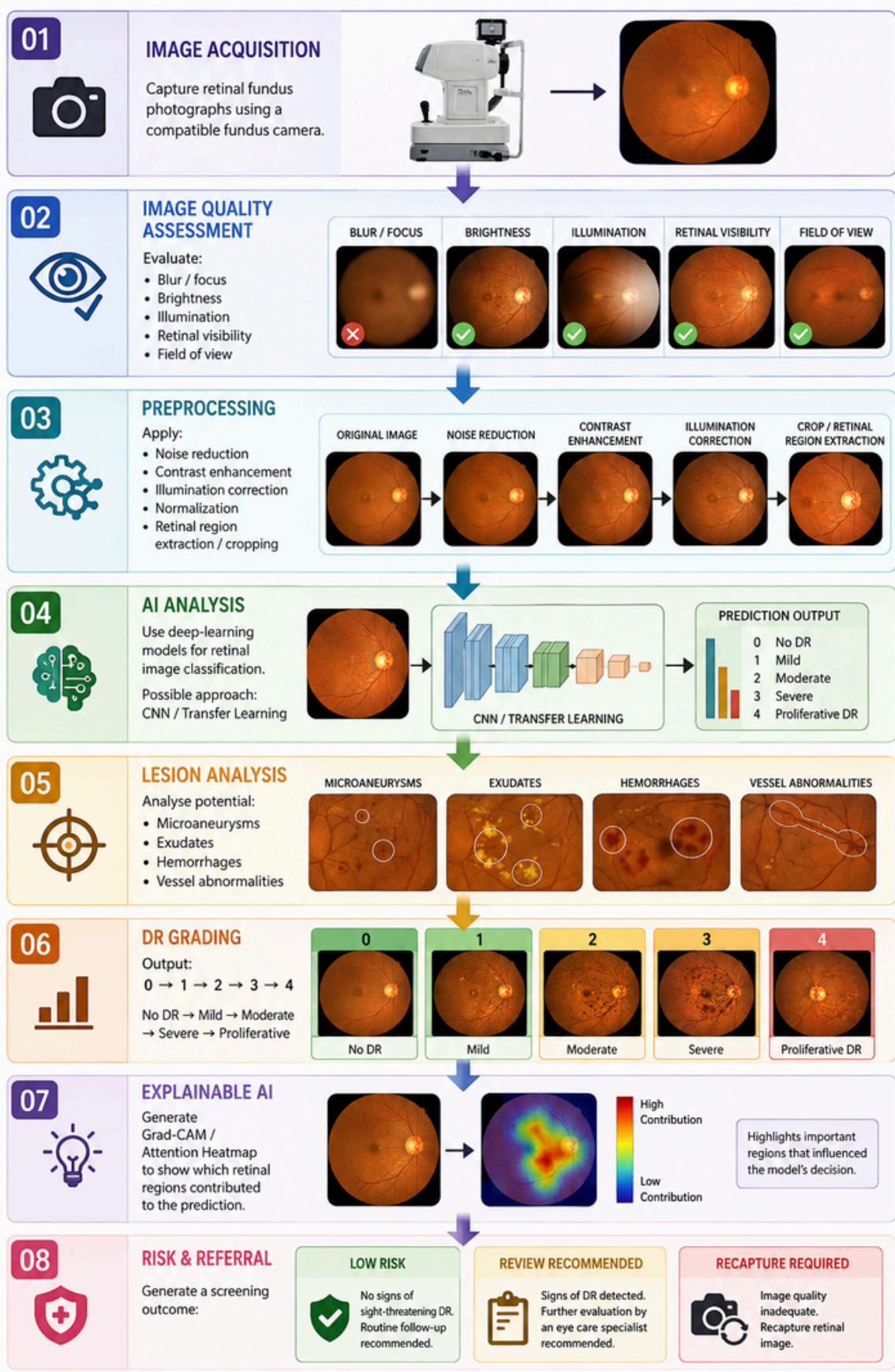
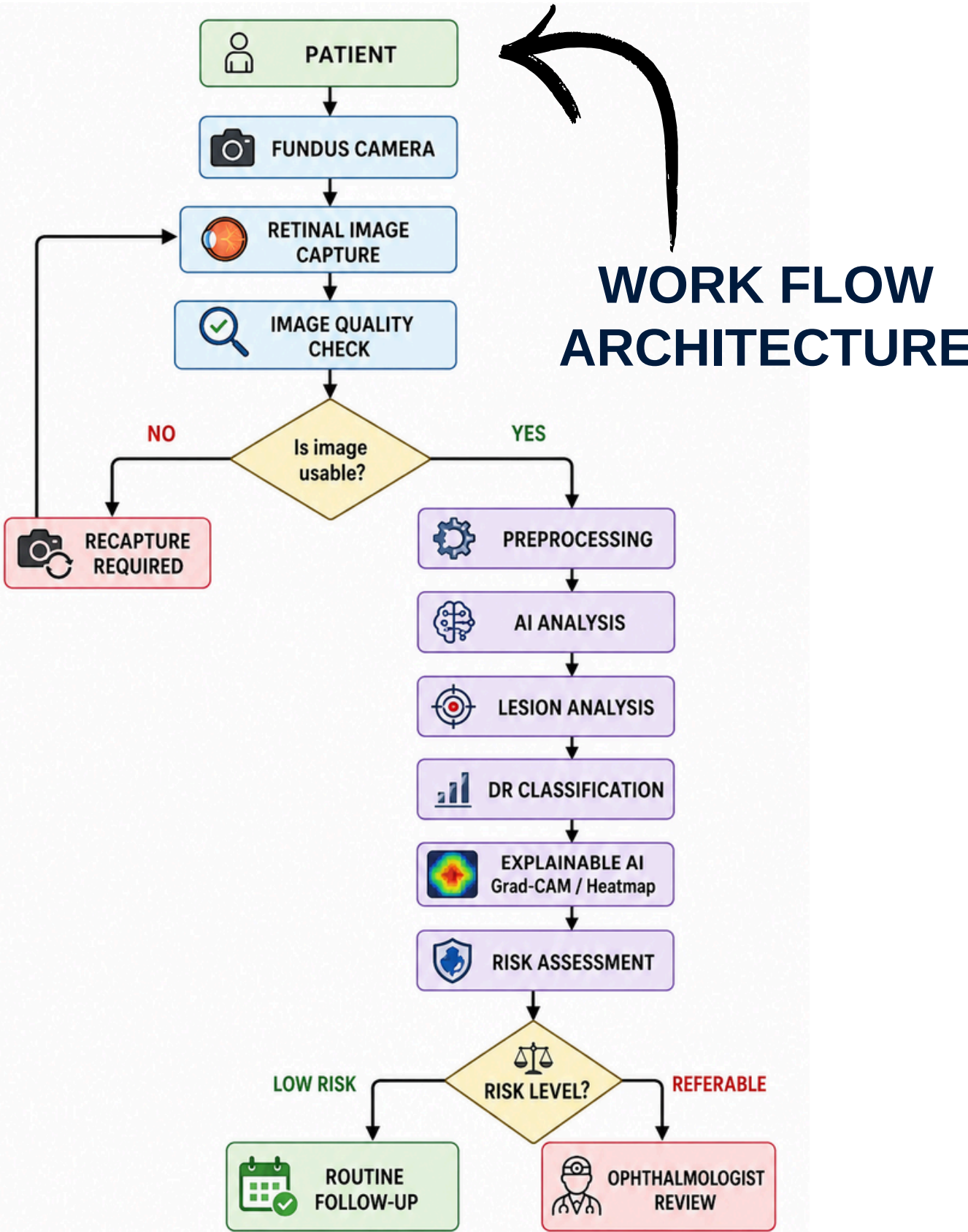
PROPOSED SOLUTION

- [AI-based DR detection](#) : automatically detects and grades diabetic retinopathy from retinal images.
- [Image-quality checking & enhancement](#) : improves poor/blurry retinal images before analysis.
- [Explainable AI \(Grad-CAM\)](#) : highlights the retinal regions responsible for the AI prediction.
- [Simulink scalability model](#) : simulates how the system can handle large-scale rural screening.

UNIQUE VALUE PROPOSITION

- [Explainable AI](#) : not just detects DR, but shows where and why it detected it.
- [Works with imperfect rural images](#) : quality checking + enhancement before diagnosis.
- [Doctor + AI collaboration](#) : AI screens and prioritizes; ophthalmologists make the final decision.
- [Scalable rural screening](#) : designed to support large-scale screening with limited specialists.







FEASIBILITY

BUILT ON PROVEN TECHNOLOGY



Technical

Uses established AI components – Computer Vision, Image Processing, Deep Learning, Explainable AI for retinal image analysis.



Operational

PHC-compatible workflow: Fundus Camera → AI Screening → Risk Prioritization → Specialist Review.



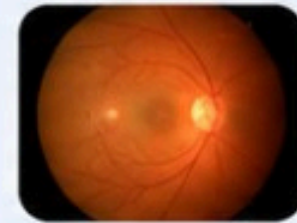
Accessibility

Can be deployed in Primary Healthcare Centres, District Hospitals, Community screening programs and Telemedicine networks.



Designed to Reduce Manual Effort

Minimizes the need for ophthalmologists to manually screen every image.



From Capture
to Analysis
in Minutes



VIABILITY

REAL USERS. REAL IMPACT. REAL OPPORTUNITIES.



Market

Potential users are Government Health Programs, Hospitals, Eye-care Networks, NGOs and Telemedicine Providers.



Revenue

Potential models:

- B2G – Government screening contracts.
- B2B – Hospital/healthcare network subscriptions.
- SaaS – Per-screening or subscription-based platform.



Scalability

AI-assisted first-level screening can process large image volumes while prioritizing suspicious cases for specialist attention.



Government Programs



Hospitals & Healthcare Networks



NGOs & Community Programs



Telemedicine Providers

SCALABLE SOLUTION FOR A HEALTHIER INDIA



TECHNICAL CHALLENGES

IDENTIFY. MITIGATE. IMPROVE.



Risk

Poor-quality retinal images caused by incorrect image capture (blur, low light, uneven illumination) may reduce AI reliability.



Mitigation

Automatically evaluates image quality, applies suitable enhancement techniques and prompts for recapture if needed, reducing screening errors.



Better
Image Quality
=
Higher
Reliability



BUSINESS / USER CHALLENGES

BUILD TRUST. ENABLE ADOPTION.



Risk

Healthcare workers may be hesitant to rely on AI predictions if the system provides only a final diagnosis without showing why the prediction was made.



Mitigation

Grad-CAM highlights the regions and displays retinal abnormalities, providing an indication of how confident it is in its prediction.



Transparency
Builds Trust
and
Adoption



HEALTHCARE IMPACT

- Identifies patients who may require specialist evaluation.
- Allows ophthalmologists to focus attention on suspicious cases.
- Supports screening closer to underserved communities.

**Earlier Detection
Better Lives**



ECONOMIC IMPACT

- Performs first level image analysis before specialist review.
- Supports large screening volumes without proportional increase in specialist screening time.
- Suspicious cases can be escalated to remote specialists.

**Cost-Effective
Scalable
Efficient**



SOCIAL IMPACT

- Brings preliminary retinal screening closer to communities with limited specialist access.
- Encourages diabetic patients to undergo regular eye screening.
- Supports earlier identification and referral rather than waiting for vision threatening symptoms.

**Accessible
Inclusive
Healthier
Communities**

Aspect	Existing Systems	LumoraVision
Process Flow	Image → Prediction	Image → Quality Check → Analysis → Explanation → Referral
Image Quality	May assume good image quality	Quality-aware screening (handles blur, low light, artifacts)
Focus	Prediction-focused	Lesion-aware + prediction
Output	Black-box output possible	Explainable AI (Grad-CAM / visual explanation)
AI Result	AI result	AI + specialist workflow
Model Usage	Screening model	Screening + triage concept
Scalability	Model performance focus	Model + healthcare workflow scalability



HOW LUMORAVISION WORKS

FROM IMAGE TO IMPACT



01 MODE-1: CLEAR IMAGE

GOOD QUALITY



For : Good quality retinal image



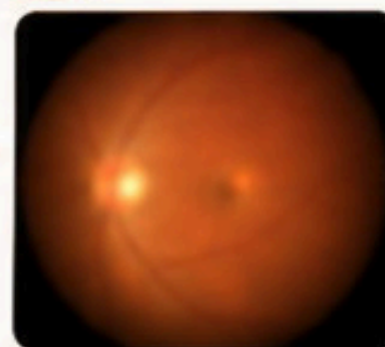
Process : Capture → Quality Pass → AI analysis → DR Grade → Explainable report



Best For : Routine diabetic retinopathy screening

02 MODE-2: BORDERLINE IMAGE

MODERATE QUALITY



For : Image has moderate blur, low contrast or uneven illumination



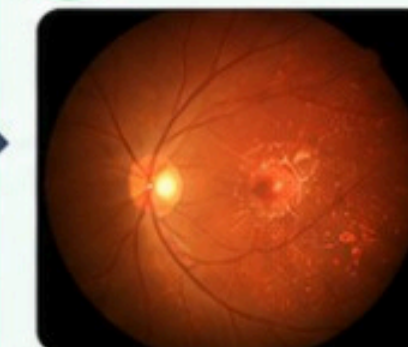
Process : Capture → Quality check → Image Enhancement → Re-analysis



Best For : Variable field / portable camera condition

03 MODE-3: REFERABLE RISK

SUSPICIOUS / HIGH RISK



For : AI identifies features requiring specialist evaluation



Process : Capture → AI Analysis → Lesion Evidence → Risk Flag → Ophthalmologist Review



Best For : Patients requiring further clinical assignment

Research source : International Clinical Classification System for Diabetic Retinopathy and Diabetic Macular Edema - 2012

<https://www.aao.org/education/clinical-statement/international-clinical-classification-system-diabe>

DATASETS: EyePACS , Messidor , APTOS

RESEARCH AREAS: Deep learning for DR , fundus image classification , Retinal lesion detection , Explainable AI/Grad-CAM

EVALUTION: *Sensitivity* - Specificity , AUROC , F1 Score , Confusion Matrix , Calibration

SOLUTIONS ALREADY EXIST

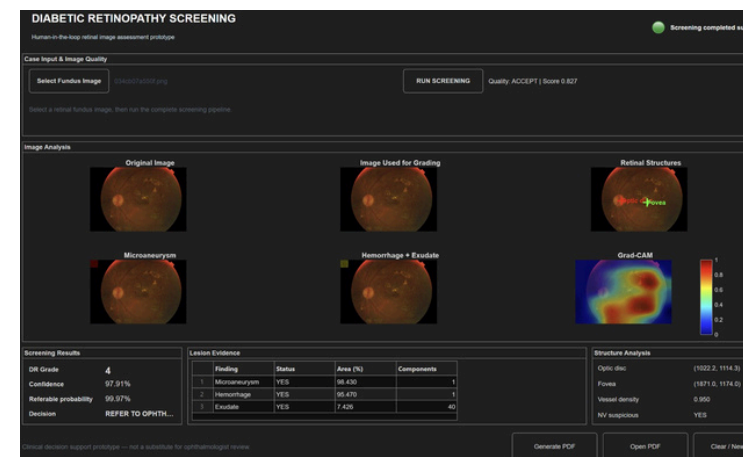
- .Automated DR Screening System.
- .Fundus Image Classification Models
- .Tele-Ophthalmology Platforms
- .Commercial AI Screening Systems



OUR SOLUTIONS STANDS OUT

- **One click:** Quick and easy retinal screening
- **AI-Powered:** Detects diabetic retinopathy early
- **Explainable:** Shows affected retinol regions
- **Accurate:** High sensitivity screening support

Demo Prototype



Prototype Report

